

B.4 Thermodynamics [HL only]

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

Use $Q = \Delta U + W$, where W is the work done *by* the gas. Show your working.

Question 1 — The espresso piston

(7 marks)

The gas in a barista-machine cylinder absorbs 100 J of thermal energy while doing 40 J of work pushing its piston outwards.

- (a) State the first law of thermodynamics, and define the sign of each term in the convention above. [3]
- (b) Calculate the change in internal energy of the gas. [2]
- (c) In the next stroke, 60 J of work is done *on* the gas while it loses 100 J of heat. Calculate ΔU for this stroke. [2]

Question 2 — The bike pump gets hot

(7 marks)

Pumping a tyre quickly compresses the air in the pump barrel so fast that essentially no thermal energy escapes during each stroke.

- (a) Name this type of process, and state the value of Q for it. [2]
- (b) Use the first law to explain why the air's temperature rises. [3]
- (c) A classmate objects: "no heat was added, so the temperature cannot change." Name the two distinct process types the classmate has merged into one. [2]

Question 3 — The optimistic inventor

(6 marks)

An inventor announces an engine operating between a boiler at 500 K and cooling water at 300 K, claiming 55% efficiency.

- (a) Calculate the maximum (Carnot) efficiency possible between these temperatures. [2]
- (b) Judge the inventor's claim. [1]
- (c) The inventor tries again with the cold reservoir at 27 °C, computing $1 - 27/500 = 95\%$. Identify the error and give the correct ceiling. [3]

Question 4 — The tidy bedroom theorem

(5 marks)

- (a) State the second law of thermodynamics in terms of entropy. [2]
- (b) When a glass of ice melts in a warm room, the ice's entropy increases. Explain, in terms of energy spreading or the number of ways of arranging energy (microstates), why the process is irreversible overall. [3]

Answers

Question 1 — The espresso piston

- (a) $Q = \Delta U + W$. In this convention: $Q > 0$ means thermal energy flows *into* the gas; $W > 0$ means work is done *by* the gas, as in an expansion; and $\Delta U > 0$ means the internal energy, and so the temperature, rises.
- (b) $\Delta U = Q - W = 100 - 40 = +60 \text{ J}$.
- (c) Losing heat makes $Q = -100 \text{ J}$, and work done *on* the gas makes $W = -60 \text{ J}$. So $\Delta U = -100 - (-60) = -40 \text{ J}$: the internal energy falls.

Question 2 — The bike pump gets hot

- (a) Adiabatic, and $Q = 0$.
- (b) With $Q = 0$ the first law gives $\Delta U = -W$. The gas is compressed, so the work done *by* the gas is negative, making ΔU positive. The internal energy therefore rises, and since internal energy tracks average molecular kinetic energy, the temperature rises. Physically, the work done on the gas by the piston becomes molecular kinetic energy.
- (c) Adiabatic ($Q = 0$) and isothermal (T constant). These are different processes: the classmate has assumed that no heat flow implies no temperature change, but in an adiabatic process the temperature changes precisely *because* work is done instead.

Question 3 — The optimistic inventor

- (a) $\eta_{\max} = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{500} = 0.40$, that is 40%.
- (b) The claimed 55% exceeds the Carnot ceiling of 40%, so it is impossible. Reject the claim.
- (c) Celsius has been fed into a formula that requires absolute temperatures: $1 - 27/500 = 95\%$ is meaningless. Converting, $27^\circ\text{C} = 300 \text{ K}$, which is the same cold reservoir as before, so the ceiling is unchanged at $1 - 300/500 = 40\%$.

Question 4 — The tidy bedroom theorem

- (a) The total entropy of an isolated system never decreases: it stays constant for a reversible process and increases for any real one.
- (b) Melting gives the water molecules access to far more arrangements than the ordered crystal allowed, and energy from the warm room spreads out into the ice. The reverse process would require energy to re-concentrate spontaneously and the molecules to re-order into a crystal while the room warmed, which corresponds to a vastly smaller number of microstates. It is not forbidden by energy conservation, merely overwhelmingly improbable, and that is what makes the melting irreversible.